Build a prediction one coordinate at a time

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Reference frame / ES: sistema de referencia / PT: referencial

Keep the origin and axes fixed.

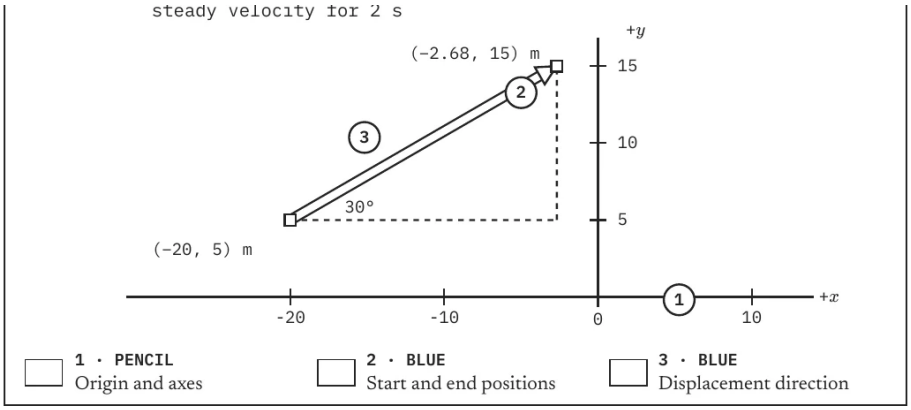
Also on your note page: Components, Units, Limit.

B READ THE RELATIONSHIP

Advance x and y independently using vx and vy in the same frame.

As you read, find: origin · direction · assumption.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A probe starts at (−20, 5) m and moves at 10 m/s , 30° from + x toward + y , for 2 s . Find its new coordinates.

Given x 0 = −20 m ; y 0 = 5 m v = 10 m/s ; θ = 30° t = 2 s ; steady v

Rule Resolve v : v x = 8.66 m/s v y = 5 m/s

Work x f = −20 + 8.66(2) y f = 5 + 5(2)

Answer (−2.68, 15) m Final x is still negative.

E SAVE FOR THE FINAL MEETING

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

Read the physics: Reference frame

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Name a reference point, mark the origin and choose axes. Record the starting position and the time. Include a unit with each value.

02 HOW IT WORKS

If velocity is steady, add velocity times elapsed time to the starting coordinate. In two dimensions, do that once for x and once for y. Use the matching component each time.

03 WHAT TO CHECK

Check whether the final position is on the expected side of zero. Say which motion assumption could fail when a new sighting arrives.

04 TRY THE EXAMPLE FROM YOUR NOTES

A probe starts at (−20, 5) m and moves at 10 m/s , 30° from + x toward + y , for 2 s . Find its new coordinates. (−2.68, 15) m Final x is still negative.

Model check: Advance x and y independently using vx and vy in the same frame.

05 CONNECT TO XJ

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

SAY IT BEFORE YOU WRITE

Would moving zero change the motion or only its coordinates? From ___ as zero, I predict ___ at time ___. I assumed ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Reference frame / ES: sistema de referencia / PT: referencial

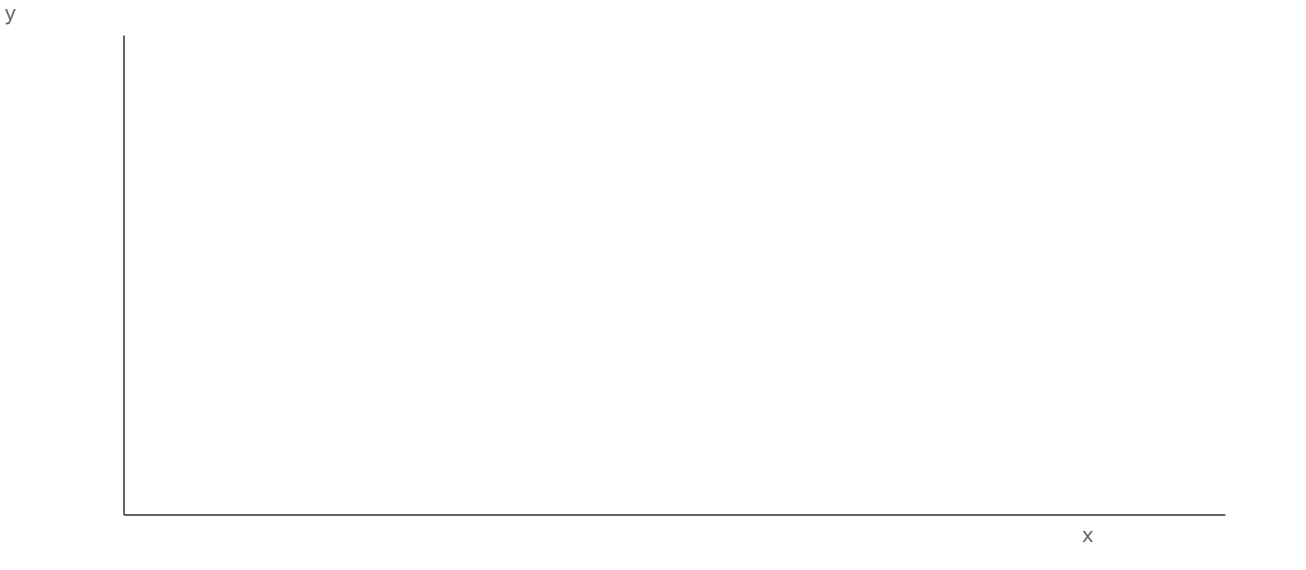
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, reference frame means …	You report a position of +5 m. What does another observer need to locate it? They need to know where ___ is and which way ___ points.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch the reference frame and one sideways component. Box your time unit and one model assumption.

Labels you can use: origin · direction · assumption.



Tell a partner: From ___ as zero, I predict ___ at time ___. I assumed ___.

On assignment: Reference frame

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two observatories file the same sighting: “5 km right of station A” and “2 km left of station B.” B is 7 km right of A.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Survey engineer**

Put both stations on one axis. Give dispatch one position in A’s frame, then list what else a future prediction needs.

☐ **Map broadcaster**

Make an on-air locator map. Convince the producer whether the two reports disagree by showing both zeros.

Word bank: origin · direction · assumption.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Name a bound that should travel with your central forecast.